NAMAN JAIN

Software Engineering | Distributed Systems | AI Applications

+91 7490994124 | naman.jain.udp04@gmail.com | LinkedIn | Github | Leetcode | Portfolio

SUMMARY

Final-year Computer Science student (CGPA 9.85/10) building scalable distributed and AI-powered systems. Experienced in backend engineering, real-time applications, and performance optimization with strong foundations in DSA, System Design, OS, DBMS, and Computer Networks.

EDUCATION

Marwadi University — B.Tech Computer Science (2023 – 2027) CGPA: **9.85**/10.00

TECHNICAL SKILLS

- **Languages:** C++, Java, Python, JavaScript, TypeScript
- **Backend & Systems:** Node.js, Express.js, FastAPI, REST APIs, WebSockets
- **Frontend:** React, Next.js, Tailwind CSS
- **Databases:** PostgreSQL, MongoDB, Prisma, Redis
- **DevOps & Cloud:** Docker, Kubernetes, AWS, GitHub Actions, CI/CD
- **AI & ML:** OpenAI API, Gemini API, Federated Learning, Computer Vision
- **CS Fundamentals:** Data Structures & Algorithms, OS, DBMS, CN, System Design

PROJECTS

Snap Attend — AI Attendance System Github Link | Live Link

- Engineered a **microservice**-based attendance platform using Next.js, FastAPI, PostgreSQL, and Docker, separating application and ML inference workloads.
- Built face-recognition attendance workflows enabling real-time identification and automated attendance tracking.
- Containerized services with **Docker** and optimized image-processing pipelines to improve reliability and scalability.

MeetAI — Real-Time AI Collaboration Platform Github Link | Live Link

- Built a real-time collaboration platform using Next.js, WebRTC, WebSockets, and OpenAI APIs supporting concurrent user sessions.
- Developed AI-powered transcription and summarization workflows, reducing manual note-taking effort by **~60%**.
- Designed secure authentication and integrated Polar.sh subscription workflows for scalable monetization.

Federated Learning System Github Link

- Implemented a privacy-preserving federated learning pipeline for distributed model training across decentralized clients.
- Compared FedAvg and FedProx optimization strategies on non-IID datasets to evaluate convergence and communication efficiency.
- Trained MobileNetV2 on ~15,000 medical images, achieving **98.93% classification accuracy**.
- Research publication in preparation on privacy-performance tradeoffs in federated learning systems.

Secure RBAC Backend System Github Link | Live Link

- Developed a secure backend system using Node.js, MongoDB, Redis, JWT, and RBAC.
- Reduced API response latency from **~2s to ~600ms** through Redis **caching** and query optimization.
- Implemented authentication, authorization, password hashing, and rate-limiting middleware to strengthen API security.
- Validated system robustness through automated Postman testing and load simulations for rate-limited endpoints.

ACHIEVEMENTS

- Solved 165+ LeetCode problems and earned the 50-Day LeetCode Badge.
- National and State-Level UCMAS Winner, solving 200 arithmetic problems in 8 minutes.
- Winner of 2 technical competitions demonstrating engineering problem-solving.
- Contributed code and fixes to ToolJet's open-source codebase during Hacktoberfest 2024.
- HackerRank Gold 5-star Java, 1-star C++, and 1-star Problem Solving badges. – HackerRank Link
- Pseudo Core Committee Member, Competitive Programming Club.